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Specialization: Metallurgical Process Engg.

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Dual Degree (B.Tech. + M.Tech.)
Gender: Female
DOB: 07/05/2003

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2025	

PROFESSIONAL EXPERIENCE

Research Intern-AI | IBM Research Labs, India (May '24 - Aug '24)

- API Sequencing and Glue Code Generation:** Utilized the IBM Granite model to generate Python main function code from user instructions, orchestrating the sequence and managing input-output relationships of existing **API helper functions**
- Created a custom dataset by Implementing a filtration process to extract and reorganize Python functions and enriched the dataset by generating detailed docstrings and task instructions using the **mistralai/mixtral-8x7b-instruct-v01** model.
- Fine-tuned the **IBM Granite** model with **QLoRA** on a created dataset, optimizing its performance to generate executable Python main function code

KEY PROJECTS

12-Lead ECG Classification Advancement Using Vision Transformer (Nov '23)

Course Project | Advance Machine Learning | Guide : Prof. Amith Sethi, Department of Electrical Engineering

- Developed and trained a **Vision Transformer** model using TensorFlow on the **PTB-XL** dataset for accurate ECG classification
- Attained an accuracy of **73.09%** and an AUC-ROC score of **74.22%**, demonstrating strong performance in ECG classification.
- Created visualizations to interpret model attention on ECG signals, enhancing understanding of the model interpretability.

GAT For Node Classification | Graph Machine Learning | Seasons Of Code | IIT Bombay (Summer '23)

- Applied **transductive** learning techniques using **Graph Attention Networks (GAT)** for node classification on Cora dataset.
- Achieved a validation accuracy of **79%** and a test accuracy of **81%** by utilizing the **negative log-likelihood** loss function.
- Visualized node embeddings from the GAT model using **t-SNE**, which condensed complex high-dimensional data into a 2D.

Image Caption Generator | Self Project | Machine Learning (Feb '23)

- Generated a **sequence of descriptive words** capturing ongoing activities within an image, prioritizing high-precision syntax
- Utilized **Inceptionv3** to extract image features, which then fed into an **LSTM** model and trained using **Flickr30k** dataset.
- Showcased the model's capacity to generate coherent and contextually appropriate captions for a wide range of images.

Sentiment Analysis with BERT | NLP & LLM's | Learner's Space | IIT Bombay (Summer'23)

- Trained a sentiment analysis model on the **SMILE Twitter** dataset, creating comprehensive training and validation sets.
- Fine-tuned a **BERT-based transformers** model using **Hugging Face**, enhancing its accuracy through multiple training epochs
- Developed an interactive demo app by **Gradio** and a saved model for real-time sentiment prediction from user input text.

Logistic Regression From Scratch | Python for Data Science | Learner's Space | IIT Bombay (Jul'23)

- Developed a logistic regression class from scratch for binary classification tasks using **Numpy**, **Pandas**, and **Matplotlib**.
- Successfully applied this model to the diabetes dataset, achieving an accuracy of **0.83** through the use of **L2 regularization**.

IPL Score Prediction | Self Project | Machine Learning (Nov '22)

- Processed IPL player statistics using **Pandas**, applied feature encoding, normalized data with **Scikit-learn's MinMaxScaler**
- Developed a **TensorFlow-Keras** neural network incorporating dropout and early stopping techniques to prevent overfitting.
- Evaluated model performance with metrics including **Mean Absolute Error (MAE)** of **9.207** and **Mean Squared Error (MSE)** of **11.2**, and gained insights by comparing predicted values against actual results.

POSITION OF RESPONSIBILITY

WiDS Project Mentor | Analytics Club, IIT Bombay (Dec '23 - Jan '24)

- Mentored 10 undergraduates on **MNIST classification with VGGNet CNN architecture**, introducing them to deep learning.
- Monitored and guided mentees through the fundamentals of **convolutional neural networks** with hands-on mini-projects.
- Advised students on optimizing VGGNet, enhancing accuracy, and understanding hyperparameter tuning and evaluation.

TECHNICAL SKILLS

Languages	:	C/C++, Python ,MATLAB, Octave, $\LaTeX$, R programming
Libraries	:	NumPy, Pandas, Matplotlib, Open CV, Scikit-Learn, TensorFlow, Keras, PyTorch
DevOps and Tools	:	Git, Visual Studio Code, LaTeX, AutoCAD, Microsoft Office

EXTRACURRICULARS

- Completed a year-long **National Sports Organization NSO** programme in **Sports** at IIT Bombay (2020-2021)
- Participated as a performer in Annual **InSync** dance show at IIT Bombay, showcasing my interest in dance (Mar'23)
- Successfully completed a month-long **$\LaTeX$** Boot Camp by **TSS Learners' Space**, IIT Bombay (summer'21)
- Worked in a team of 50+ members as a mentor for the **Abhyuday's College Representative** program (2021)